KAIZEN MATHS

Lesson starter

Binomial Expansion

Expansion Fundamentals

Name _____

S1

$(x+1)^4$

S2

Calculate ${}^5C_2$

S3

Expand $(x+2)^5$

S4

Expand $(x-5)^6$

Show your working

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ALG-C4F7-L1 · E

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Teacher answers & working

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S1

$(x+1)^4$

Answer:

$x^4+4x^3+6x^2+4x+1$

Working:

Method focus: use the Pascal row as the coefficients for the expansion.

Apply binomial theorem

$(x+1)^4=\sum_{r=0}^{4}{}^4C_r x^{4-r}$

Coefficients are 1, 4, 6, 4, 1

$x^4+4x^3+6x^2+4x+1$

Final answer: $x^4+4x^3+6x^2+4x+1$

S2

Calculate ${}^5C_2$

Answer:

10

Working:

Method focus: calculate nCr by cancelling factorials before multiplying.

Use the combination formula

${}^5C_2=\frac{5!}{2!(3)!}$

Cancel and multiply only the needed terms

${}^5C_2=\frac{5\times4}{2!}=\frac{20}{2}=10$

Final answer: 10

S3

Expand $(x+2)^5$

Answer:

$x^5+10x^4+40x^3+80x^2+80x+32$

Working:

Method focus: identify the two terms in the bracket and apply the binomial theorem.

Apply $(a+b)^n=\sum {}^nC_r a^{n-r}b^r$

Here $a=x, b=2, n=5$

Pascal row: 1, 5, 10, 10, 5, 1

$x^5+10x^4+40x^3+80x^2+80x+32$

Final answer: $x^5+10x^4+40x^3+80x^2+80x+32$

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S4

Expand $(x-5)^6$

Answer:

$x^6-30x^5+375x^4-2500x^3+9375x^2-18750x+15625$

Working:

Method focus: treat subtraction as adding a negative term, so signs alternate.

Rewrite as $(x+(-5))^6$

Signs alternate because $(-5)^r$

$x^6-30x^5+375x^4-2500x^3+9375x^2-18750x+15625$

Final answer: $x^6-30x^5+375x^4-2500x^3+9375x^2-18750x+15625$

E1

$(x+1)^6$

Answer:

$x^6+6x^5+15x^4+20x^3+15x^2+6x+1$

Working:

Method focus: use the Pascal row as the coefficients for the expansion.

Apply binomial theorem

$(x+1)^6=\sum_{r=0}^{6}{}^6C_r x^{6-r}$

Coefficients are 1, 6, 15, 20, 15, 6, 1

$x^6+6x^5+15x^4+20x^3+15x^2+6x+1$

Final answer: $x^6+6x^5+15x^4+20x^3+15x^2+6x+1$

E2

Calculate ${}^7C_4$

Answer:

35

Working:

Method focus: calculate nCr by cancelling factorials before multiplying.

Use the combination formula

${}^7C_4=\frac{7!}{4!(3)!}$

Cancel and multiply only the needed terms

${}^7C_4=\frac{7\times6\times5\times4}{4!}=\frac{840}{24}=35$

Final answer: 35

Teacher answers & working

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E3

Expand $(x+3)^3$

Answer:

$x^3+9x^2+27x+27$

Working:

Method focus: identify the two terms in the bracket and apply the binomial theorem.

Apply $(a+b)^n=\sum {}^nC_r a^{n-r}b^r$

Here $a=x, b=3, n=3$

Pascal row: 1, 3, 3, 1

$x^3+9x^2+27x+27$

Final answer: $x^3+9x^2+27x+27$